

```
Import numpy as np
```

```
From sklearn.cluster import KMeans
```

```
# Example list of [image directory, (3×4) numpy array]
```

```
Data = [
```

```
    ["img1.jpg", np.random.rand(3, 4)],
```

```
    ["img2.jpg", np.random.rand(3, 4)],
```

```
    ["img3.jpg", np.random.rand(3, 4)],
```

```
    ["img4.jpg", np.random.rand(3, 4)]
```

```
]
```

```
# Step 1: Stack all (3×4) matrices into a NumPy array (Shape: (N, 3, 4))
```

```
Feature_matrices = np.array([item[1] for item in data]) # Shape: (N, 3, 4)
```

```
Image_dirs = [item[0] for item in data] # Extract image directories
```

```
# Step 2: Reshape for KMeans clustering
```

```
Num_samples = feature_matrices.shape[0] # Number of images
```

```
Feature_matrices = feature_matrices.reshape(num_samples, 3, 4, 1) # Shape: (N, 3, 4, 1)
```

```
# Step 3: Apply KMeans Clustering (Using 3D data directly)
```

```
Num_clusters = 2 # Adjust based on your dataset
```

```
Kmeans = KMeans(n_clusters=num_clusters, random_state=42, n_init=10)
```

```
# Instead of flattening, use .reshape(N, -1) to keep structure while clustering
```

```
Feature_vectors = feature_matrices.reshape(num_samples, -1) # Shape: (N, 12)
```

```
Kmeans.fit(feature_vectors)
```

```
# Step 4: Get cluster assignments
```

```
Cluster_labels = kmeans.labels_
```

```
Centroids = kmeans.cluster_centers_.reshape(num_clusters, 3, 4) # Reshape back to (3,4)
```

```
# Step 5: Organize results as [centroid, image directories]
```

```
Clusters = {}
```

```
For idx, label in enumerate(cluster_labels):
```

```
    If label not in clusters:
```

```
        Clusters[label] = {"centroid": centroids[label], "image_dirs": []}
```

```
    Clusters[label]["image_dirs"].append(image_dirs[idx])
```

```
# Convert to list format
```

```
Final_result = [[v["centroid"], v["image_dirs"]] for v in clusters.values()]
```

```
# Print the clusters
```

```
For cluster in final_result:
```

```
    Print("Centroid:\n", cluster[0]) # 3x4 Matrix
```

```
    Print("Images:", cluster[1])
```

```
    Print()
```